Open in app ↗ Sign up Sign in

Medium Search Write

Member-only story

MATHEMATICAL MYSTERIES

You Can Fill It, But You Can’t Paint It

The Strange Case of Gabriel’s Horn

Vagelis Plevris Follow 8 min read · 1 day ago

233 4

There is a **peculiar trumpet** in mathematics.

It is not made of brass or wood. **It is not something you can hold or hear**. It exists only as an exact mathematical construction, defined by a simple equation. And yet, despite its simplicity, it behaves in a way that feels almost impossible.

Imagine this horn stretching out into space, longer and longer, becoming thinner and thinner, never quite ending.

Now consider this. If I gave you a gallon of paint, **you could pour it inside and fill it completely**.

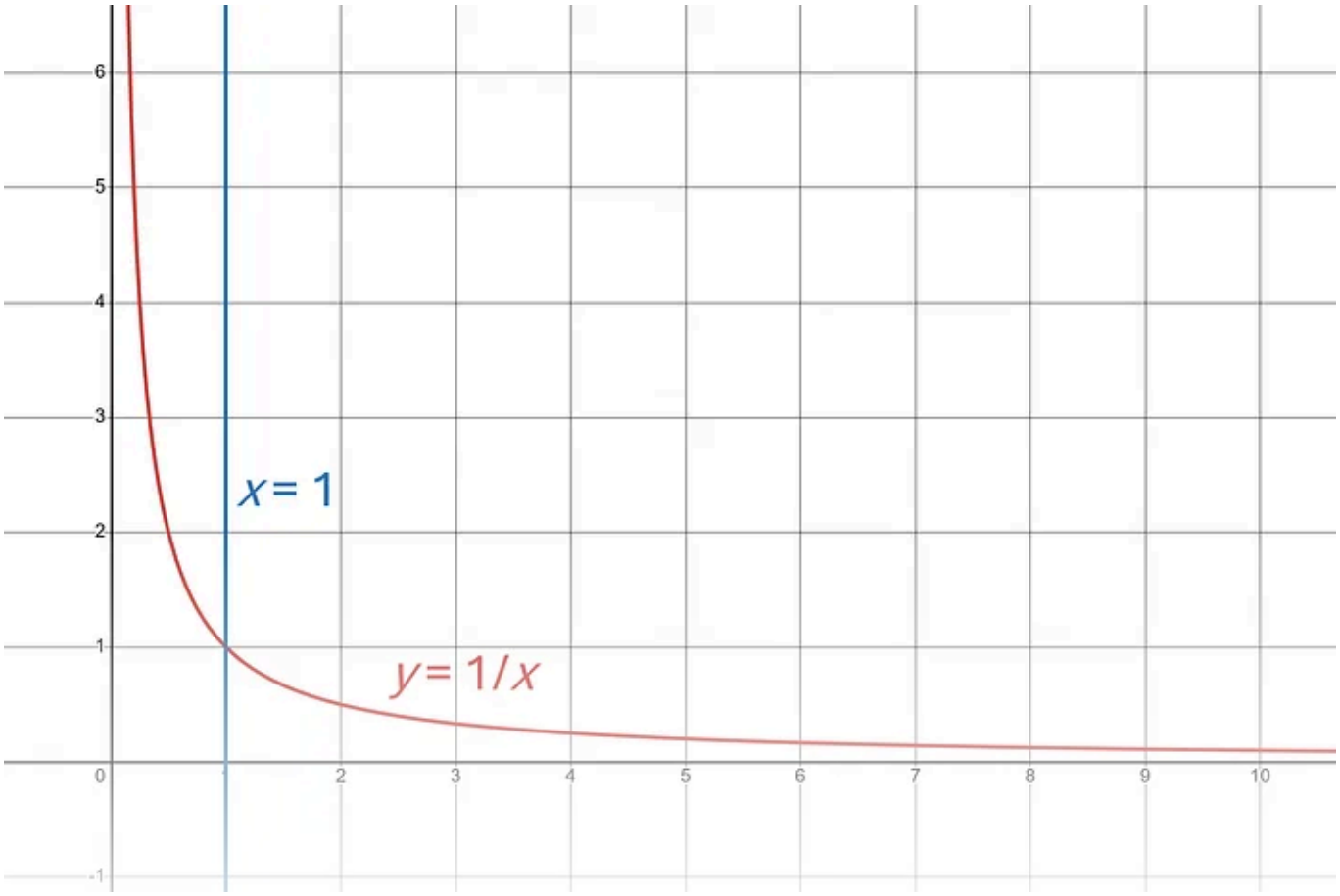
But if I asked you to paint its outer surface, **even all the paint in the world would not be enough**.

The same object can be filled with a finite amount of paint, and at the same time requires an infinite amount to cover.

That is the paradox of **Gabriel’s horn.**

The object that should not exist

We begin with a very simple graph.



Graph of y = 1/x (red) together with the vertical line x = 1 (blue). The portion of the curve for x ≥ 1 is used to generate Gabriel’s horn by rotation about the x-axis.

Create an account to read the full story.

The author made this story available to Medium members only.
If you’re new to Medium, create a new account to read this story on us.

Sign up with Google

Sign up with Facebook

Sign up with email

Already have an account? Sign in

Written by Vagelis Plevris

3K followers · 77 following

Structural engineer, professor and science communicator making complex ideas simple. Passionate about research, teaching and improving lives through knowledge.

Follow

Responses (4)

Write a response

What are your thoughts?

Americantrustfactory
10 hours ago

Beautifully crafted body of work. I enjoyed reading this. 📯

6 Reply

Abrie Nel
15 hours ago

Fascinating article, and very well written!

5 Reply

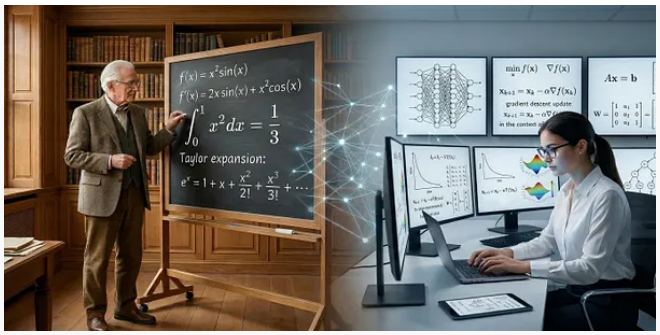
Nathanael he/him
17 hours ago

A legendary paradox!

5 Reply

See all responses

More from Vagelis Plevris

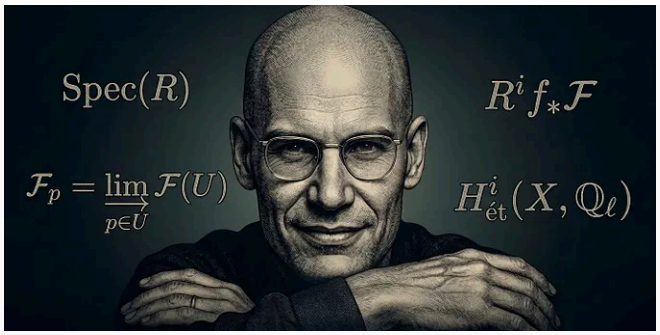


Vagelis Plevris

Mathematics Undergoing Its Biggest Transformation in History

AI can now generate proofs, explore ideas, and solve problems. What remains is decidin...

Mar 25

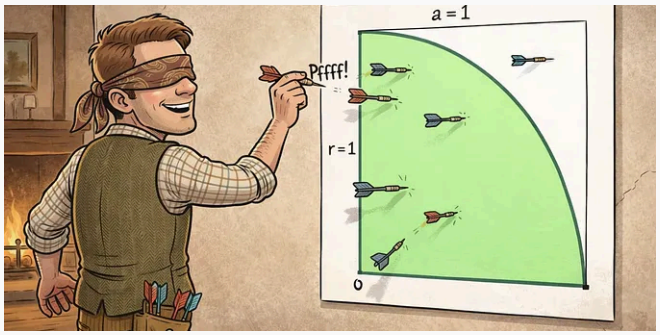


Vagelis Plevris

Alexander Grothendieck: The Man Who Rebuilt Geometry — and...

He rebuilt the foundations of modern geometry, reshaped number theory, and the...

Mar 6



Vagelis Plevris

Monte Carlo Simulation: The Method That Turned Randomnes...

How a solitaire question at Los Alamos reshaped science, finance, and artificial...

Mar 1

Vagelis Plevris

The Man Who Taught Mathematics How to Speak

How Leonhard Euler gave mathematics its modern language and kept building it even...

Feb 27

See all from Vagelis Plevris

Recommended from Medium

In New East by Yuri Minamide

Four Cool Japanese Home Appliances the World Hasn’t...

You might end up wanting one or two of them

Feb 20

In ILLUMINATION by Sufyan Maan, M.Eng

I Woke Up at 4:30 AM Every Day for 30 Days — Here Is What Nobody...

Here is what actually happened, from someone who did it & tracked everything.

Mar 8

Chris Ferrie

Quantum entanglement is far stranger than you’ve been told

Quantum entanglement. What the hell is it? Perhaps you’ve read that

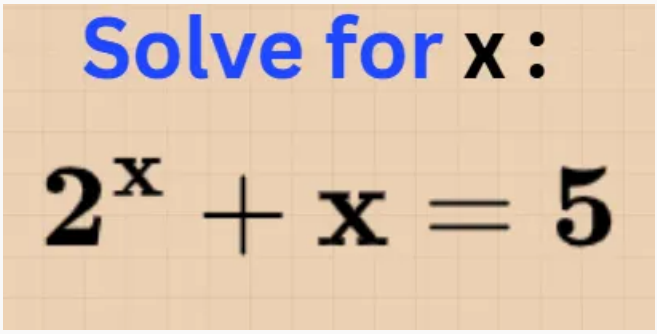
Mar 14

EdithTali

Why Men Choose Divorced Women Over “Leftover” Singles: The Trut...

Three men reveal the brutal truth about why they’d rather marry divorced women than...

Mar 2

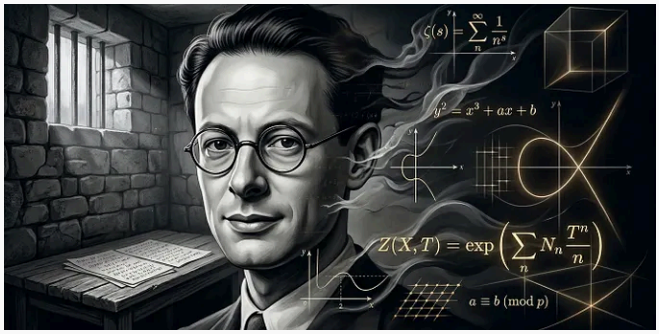


In Puzzle Sphere by Thanushan Sivapatham

A Harvard University Exam Question

An instruction in organization rather than calculation

Feb 11



Vagelis Plevris

André Weil: The Mathematician Who Proved a Theorem in Prison

How a prison sentence in 1940 led to ideas that would eventually help solve a 350-year-...

5d ago

See more recommendations